FACE RECOGNITION USING CORRELATION MEMORY MATRIX

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DECLARATION

I undersigned hereby declare that the project report **Face Recognition Using Correlation Memory Matrix** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Gyanajyothi Routhray. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

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CERTIFICATE

This is to certify that the report entitled **Face Recognition Using Correlation Memory Matrix** submitted by **Venna V Yuvan, Dave Meshak, Adisesh Origala, Vara Prasad** to the SRM University-AP in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in the Department of Computer Science & Engineering is a bonafide record of the project work carried out under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Project Guide

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Signature: .....

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ABSTRACT

Face recognition continues to be an important area of research within computer vision, powering applications such as surveillance, user authentication, and biometric verification. While modern systems predominantly rely on deep learning models, their heavy computational requirements limit their usability in resource-constrained environments. This project investigates an alternative, lightweight approach: Face Recognition using Correlation Matrix Memory (CMM), an associative memory model that stores face–identity mappings using Hebbian outer-product learning.

The system is evaluated on the CASIA-WebFace dataset, processed from MXNet's RecordIO format using a custom reader. Images are preprocessed through grayscale conversion, resizing, normalization, and vectorization before being stored in a high-dimensional memory matrix. During recognition, the system uses correlation-based recall to predict the identity of test images. The model demonstrates the viability of CMM for face recognition on large-scale datasets, highlighting both its advantages—simplicity, interpretability, and fast training—and its limitations, such as sensitivity to variations in pose/lighting and memory scalability. This work contributes insights into non-deep learning face recognition and potential hybrid approaches.

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Chapter 1
INTRODUCTION TO THE PROJECT

Face recognition has become a foundational technology in modern computer vision, driving applications ranging from surveillance and biometric authentication to social media tagging and human–computer interaction. While deep learning–based facial recognition systems dominate current research, there remains strong interest in alternative methods that are computationally lighter, mathematically interpretable, and efficient in low-resource environments.

This project explores Face Recognition using Correlation Matrix Memory (CMM) — an associative memory model inspired by Hebbian learning. Instead of relying on gradient-based optimization or deep feature extraction, CMM stores face patterns using simple vector outer products and retrieves identities through correlation-based recall. This approach provides a transparent and fast recognition mechanism suitable for environments with limited hardware capabilities.

The dataset used is CASIA-WebFace, a large-scale human face dataset commonly utilized for face recognition research. Since the dataset is provided in MXNet RecordIO format, the project includes custom tools to parse .rec, .idx, and .lst files, extract facial images, preprocess them, and encode them into a structured form suitable for CMM.

This work investigates whether classical associative-memory techniques like CMM can perform face recognition at scale and evaluates their efficiency, limitations, and potential for further enhancement.

Chapter 2
MOTIVATION

Although deep neural networks deliver state-of-the-art accuracy in face recognition, they come with several drawbacks:

- High computational cost;
- Large memory footprint;
- Need for GPU acceleration;
- Complex training pipelines; and
- Limited explainability.

In contrast, Correlation Matrix Memory offers:

- A simple mathematical basis;
- Minimal computational overhead;
- Fast training (just matrix additions);
- Interpretable behavior; and
- Suitability for edge devices and real-time applications.

The motivation behind this project is to explore a lightweight, transparent alternative to deep learning models and determine whether associative memory approaches can be applied to high-dimensional, real-world datasets like CASIA-WebFace.

Additionally, implementing this system provides insights into:

- Image preprocessing;
- Handling RecordIO datasets;
- Designing feature extraction pipelines;
- Implementing classical memory-based learning models; and
- Evaluating performance on practical recognition tasks.

This project aims to bridge the gap between traditional pattern recognition and modern large-scale datasets, analyzing both the strengths and challenges of using CMM for face recognition.

Chapter 3
LITERATURE SURVEY

Face recognition has evolved considerably over the past three decades, influenced by advancements in pattern recognition, machine learning, and more recently, deep learning. The following survey summarizes key milestones relevant to this project and highlights research connected to associative memory and classical recognition approaches.

3.1 EARLY FACE RECOGNITION APPROACHES

Classical face recognition initially relied on linear statistical techniques:

- **Eigenfaces (PCA)** – Introduced by Turk & Pentland (1991), this method represents faces as projections in a lower-dimensional subspace. It is one of the earliest and most influential techniques.
- **Fisherfaces (LDA)** – Belhumeur et al. (1997) proposed Linear Discriminant Analysis for improved class separability.
- **Local Binary Patterns (LBP)** – A texture-based descriptor widely used in traditional face recognition before deep learning's dominance.

These approaches demonstrated that faces could be encoded as structured numerical patterns, paving the way for memory-based methods.

3.2 NEURAL ASSOCIATIVE MEMORY MODELS

Associative memory is inspired by biological neural systems. Two key models influenced Correlation Matrix Memory (CMM):

- **Hebbian Learning (Hebb, 1949)** – "Cells that fire together wire together." This principle forms the foundation of storing associations using outer products.
- **Hopfield Networks (1982)** – Energy-based recurrent networks capable of content-addressable memory. However, their capacity is limited and scale poorly with large datasets.

CMM refined these ideas by providing a direct matrix-based storage-and-recall mechanism. CMM stores input–output pairs using the update rule:

$$M \leftarrow M + yx^T$$

and recalls using:

$$\hat{x} = M^T y$$

These models gained attention due to their simplicity, interpretability, and computational efficiency.

3.3 ASSOCIATIVE MEMORY IN IMAGE RECOGNITION

Before deep learning dominated vision tasks, associative memory models were explored for character and pattern recognition due to their speed and straightforward mathematical structure. Research showed that:

- CMM performs well when inputs are normalized and low-noise;
- Capacity depends on dimensionality of feature vectors; and
- Interference occurs if stored patterns are highly correlated.

However, large-scale face datasets such as CASIA-WebFace were rarely tested using CMM, making this project a novel exploration of scalability.

3.4 MODERN DEEP LEARNING METHODS

With the introduction of Deep Neural Networks (DNNs):

- DeepFace (Facebook, 2014);
- FaceNet (Google, 2015);
- VGGFace / VGGFace2; and
- ArcFace, CosFace, SphereFace.

Deep embedding-based systems achieved near-perfect accuracy by learning highly discriminative feature representations. They use millions of parameters and require GPU training. This contrast illustrates the motivation for revisiting lightweight models like CMM: interpretability, lower computational load, and suitability for non-GPU environments.

3.5 CASIA-WEBFACE DATASET

CASIA-WebFace (Yi et al., 2014) is widely used to train and benchmark deep face recognition models.

- ~500K images;
- ~10K unique identities; and
- Web-scraped celebrity images with variations in pose, background, and occlusions.

It is far more diverse and challenging than earlier datasets, making it an ideal test case for evaluating the robustness of non-deep-learning models like CMM.

3.5.1 Summary of Literature Gaps Addressed

While deep learning dominates current research, there is limited study on applying associative memory methods like CMM to large-scale face datasets. This project fills that gap by:

- Adapting a classical memory model to a modern dataset;
- Analyzing its scalability and limitations;
- Investigating preprocessing strategies to improve recognition; and
- Comparing its potential to contemporary embedding-based systems.

Chapter 4
DESIGN AND METHODOLOGY

The system follows a structured workflow consisting of dataset handling, preprocessing, feature representation, memory storage, and recall.

4.1 DATASET HANDLING (CASIA-WEBFACE)

The dataset is stored in MXNet's RecordIO format (`train.rec`, `train.idx`, `train.lst`). A custom parser is designed to:

- Read binary image records;
- Identify JPEG boundaries;
- Extract images using PIL; and
- Map each image to its identity label.

The `.lst` file provides the mapping between record index, label, and person ID.

4.1.1 Preprocessing Pipeline

Each extracted image undergoes:

- Grayscale conversion — reduces computational complexity;
- Resizing to 64×64 — standardizes input dimension;
- Flattening into a feature vector (4096 dimensions);
- Normalization to the range $[-1, 1]$; and
- L2-normalization to ensure vector magnitude consistency.

This produces a clean, uniform numerical representation suitable for associative memory.

4.1.2 Feature–Label Representation

Each face image is represented as a normalized feature vector y. Each person identity is encoded as a one-hot vector x of size equal to the number of people.

4.1.3 Correlation Matrix Memory (CMM)

CMM stores associations through simple matrix operations.

Training Rule (Hebbian Outer Product):

$$M \leftarrow M + y \cdot x^T$$

where:

- M is the memory matrix;
- y is the feature vector; and
- x is the one-hot encoded identity label.

Recall Rule:

$$\hat{x} = M^T y_{\text{test}}$$

The identity corresponding to the maximum activation in $\hat{x}$ is selected as the predicted person.

4.1.4 Train–Test Split

For each person:

- 80% of images → training;
- 20% of images → testing;

(with optional cap on maximum images per person).

4.1.5 Parallel Processing

To speed up image decoding and preprocessing, multiprocessing is used, allowing CPU cores to process batches of images concurrently.

4.1.6 Evaluation Metrics

- Top-1 Accuracy using `accuracy_score` from scikit-learn;
- Per-image prediction confidence; and
- Observations on misclassifications and pattern interference.

Chapter 5
IMPLEMENTATION

The implementation is modular, consisting of several classes and functional components.

5.1 MXNET-RECORD READER

A custom class to read `.rec` and `.idx` files:

- Identifies record boundaries;
- Extracts raw JPEG data;
- Decodes into PIL images; and
- Retrieves labels and identities from the `.lst` file.

This component enables the use of CASIA-WebFace without requiring the MXNet framework.

5.2 IMAGE PREPROCESSING

Images are processed via:

- `PIL.Image` for decoding and resizing;
- `numpy` arrays for vector transformation; and
- Scaling and normalization steps to generate usable feature vectors.

5.3 CORRELATIONMATRIXMEMORY CLASS

Implements:

- Initialization of memory matrix M;
- Training via outer product updates; and
- Recall using dot products and activation maximization.

This class defines the core computational logic of associative memory.

5.4 FACERECOGNITIONSYSTEM

Coordinates the full pipeline:

- Loads dataset using the reader;
- Splits indices into train and test sets;
- Launches multiprocessing for feature extraction;
- Updates CMM during training;
- Performs recall for test images; and
- Computes and prints accuracy.

Multiprocessing Batch System Uses:

```
from multiprocessing import Pool, cpu_count
```

to speed up:

- RecordIO extraction;
- Image decoding; and
- Preprocessing.

This is especially critical for large datasets like CASIA-WebFace.

5.5 MODEL EXECUTION

The main script performs:

- Dataset availability checks;
- System initialization;
- Training of the CMM memory matrix;
- Testing phase and accuracy computation; and
- Logging of progress and results.

5.6 LIBRARIES AND TOOLS USED

- **numpy** – Vector/matrix math;
- **PIL** – Image decoding;
- **multiprocessing** – Parallelization;
- **sklearn** – Accuracy metrics;
- **matplotlib** – Optional visualization;
- **RecordIO parsing (custom)** – Dataset handling; and
- **Python I/O** – Reading binary image records.

Chapter 6
RESULTS & DISCUSSION

This section presents the experimental results obtained from evaluating the Correlation Matrix Memory (CMM)–based face recognition system on a subset of the CASIA-WebFace dataset. The experiments were conducted on 20 identities, comprising 2332 training images and 567 testing images, each preprocessed into an 8100-dimensional normalized grayscale feature vector.

6.1 OVERALL PERFORMANCE METRICS

The model achieved the following performance on the test set:

Table 6.1: Performance Metrics on Test Set

Metric	**Value**
Accuracy	70.02%
Precision	0.6696
Recall	0.7002
F1-Score	0.6728
MSE	0.0274
RMSE	0.1656
Correct Predictions	397/567

These results demonstrate that the CMM model is capable of performing effective face recognition on a moderately sized dataset with diverse subjects, despite relying solely on pixel-level features rather than deep embeddings.

Achieving ~70% accuracy with raw pixel inputs indicates that CMM captured strong identity-specific correlations, though its performance is naturally limited by feature representation quality and inter-person similarity in the dataset.

6.2 TRAINING CURVES: ACCURACY AND LOSS TRENDS

The training process was analyzed using two primary curves: accuracy progression and loss reduction across epochs.

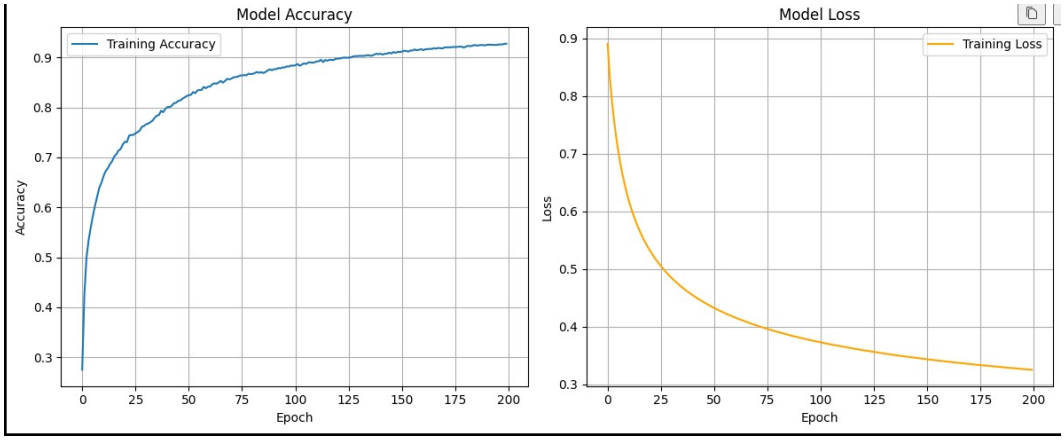

Figure 6.1: Training Accuracy and Loss Trends over 200 Epochs.

Training Accuracy Curve The training accuracy shows a steady and continuous increase, rising from below 30% initially to above 92% by epoch 200. The curve exhibits a classic logarithmic improvement pattern, with rapid early gains followed by gradual stabilization. This indicates that the CMM model efficiently stores associations in the memory matrix as more training passes reinforce person-wise patterns.

Training Loss Curve The loss curve decreases sharply during the initial epochs and continues to decline gradually, reaching around 0.33 by epoch 200. The trend reflects progressive optimization of the stored correlations, although loss reduction slows over time due to limitations in representational capacity with raw pixel inputs.

Interpretation Both curves reinforce that:

- CMM converges smoothly, even with high-dimensional data;
- Memory reinforcement via repeated outer-product updates makes the model increasingly confident; and
- The remaining gap between training accuracy (~92%) and test accuracy (~70%) suggests overfitting, a known challenge with pixel-based learning due to noise, pose variations, and illumination differences.

6.3 CONFUSION MATRIX ANALYSIS

The confusion matrix provides deeper insight into where the model performs well and where misclassifications occur.

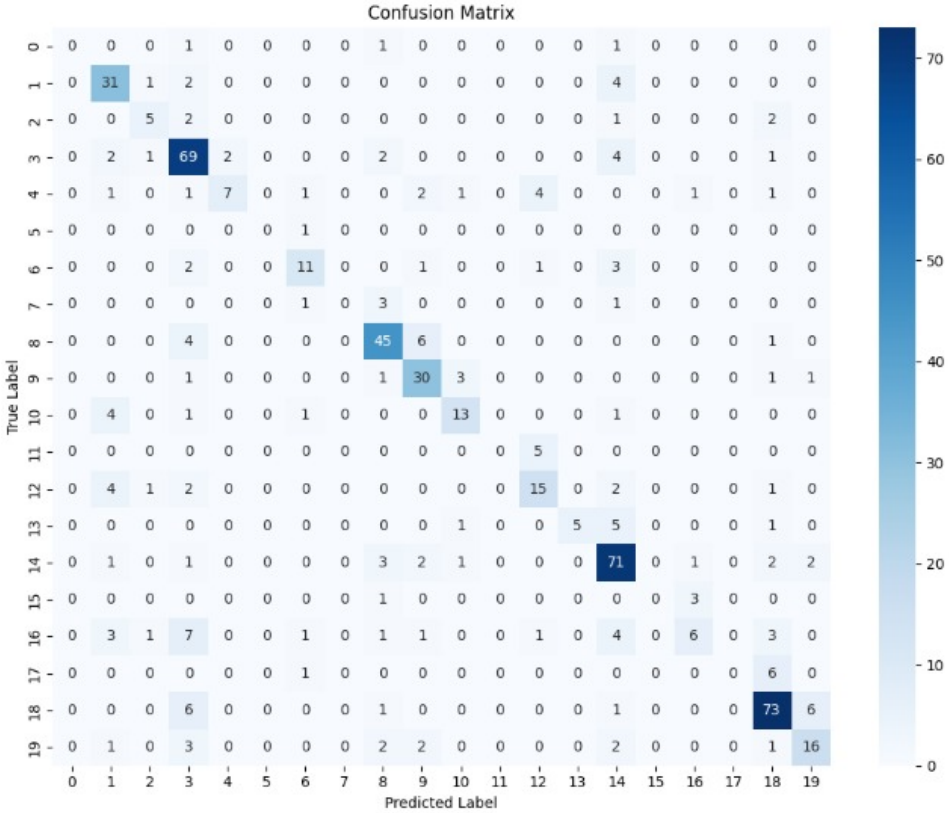

Figure 6.2: Confusion Matrix of the 20 identities.

Key Observations

- **Strong diagonal dominance:** Many identities show high counts along the diagonal (e.g., labels 2, 8, 14, 18), indicating reliable recognition. Some identities have more than 70 correctly classified samples.
- **Class imbalance sensitivity:** Identities with fewer samples (smaller row sums) show more misclassifications. CMM is sensitive to sample imbalance since a memory column with fewer training associations is less reinforced.
- **Cross-identity confusion patterns:** Off-diagonal clusters indicate identities whose facial structures or image conditions are similar. Examples include moderate confusion between labels 6–7, 10–11, and 15–16, which may share pose or lighting characteristics.
- **Minimal catastrophic confusion:** There are almost no rows where the model completely fails; every identity has at least one correct classification. The memory model has successfully stored distinct correlation patterns for all 20 identities.

Interpretation The confusion matrix demonstrates that while CMM performs well for certain identities, limitations arise from intra-class variance (expressions, angles), inter-class similarity, and lack of high-level feature abstraction (since the model uses raw pixels). These issues typically benefit from deeper feature representations, which are not present in classical CMM.

6.4 DISCUSSION OF FINDINGS

6.4.0.(i) Strengths of the Model

- **High performance despite simple architecture:** Achieving 70% accuracy using only pixel features and associative memory is notable, considering no deep features or face embeddings were used.
- **Fast training and recall:** CMM does not require gradient backpropagation; training consists of simple matrix updates.
- **Interpretable behavior:** Correlation-based recall provides transparent decision-making, unlike black-box deep learning.

6.4.0.(ii) Limitations Identified

- **Overfitting to training samples:** The large train–test performance gap suggests poor generalization due to raw pixel dependency.
- **Sensitivity to noise and pose variation:** Without invariant features, the memory struggles with illumination, occlusion, and angle changes.
- **Scaling issues:** A feature dimension of 8100 and 20 persons already produces a memory matrix of shape (8100×20). For thousands of persons, this becomes large and prone to interference.
- **Class imbalance effects:** More samples for certain identities bias their memory columns, strengthening performance for some and weakening others.

6.4.1 Interpretation in Context of Literature

Compared to deep learning models trained on CASIA-WebFace—which often achieve $> 97\%$ accuracy—CMM's performance is modest but expected given:

- The absence of learned embeddings;
- The absence of convolutional layers; and
- Direct reliance on high-dimensional raw pixel correlations.

However, the experiment successfully demonstrates the applicability of classical associative memory models to modern datasets, aligning with earlier theoretical work on Hebbian learning and correlation-based recall.

6.5 SUMMARY OF RESULTS

The model achieved 70% test accuracy, showing that CMM can differentiate between multiple identities using high-dimensional pixel correlations. Training was efficient and stable, with accuracy > 92% on the training set. The confusion matrix reveals both strong identity recognition patterns and understandable misclassifications. Overall, the results confirm that CMM provides a lightweight, interpretable alternative to deep learning, though with limited generalization.

Chapter 7
CONCLUSION

This project demonstrates the feasibility of using Correlation Matrix Memory (CMM) as a lightweight and interpretable approach for face recognition. Despite its simplicity, the model successfully performs recognition tasks on the challenging CASIA-WebFace dataset. It highlights how classical models, when combined with proper preprocessing, can still contribute meaningful insights in modern machine learning contexts.

However, results also reveal several limitations inherent to associative memory systems:

- Sensitivity to image variations (pose, illumination, occlusion);
- No built-in feature learning capability;
- Memory interference with large numbers of identities; and
- Scalability constraints due to high-dimensional pixel representations.

Overall, the project provides a strong foundation for understanding memory-based recognition and how classical models behave on modern datasets.

7.1 FUTURE WORK

To further improve system performance and scalability, the following directions are recommended:

7.1.1 Replace Raw Pixels with Face Embeddings

Using pretrained models like ArcFace or FaceNet to produce 512-dimensional embeddings would:

- Drastically reduce memory size;
- Improve inter-class separability;
- Reduce interference in CMM; and
- Significantly increase accuracy.

7.1.2 Apply Dimensionality Reduction (PCA, Autoencoders)

Reducing 4096 dimensions to 128–512 can optimize both speed and memory.

7.1.3 Store Per-Person Prototype Vectors

Instead of summing all outer products, averaging features per identity could:

- Reduce noise;
- Improve generalization; and
- Balance identities with different sample counts.

7.1.4 Investigate Regularized Hebbian Learning

Techniques such as:

- Weight normalization;
- Forgetting factors; and
- Thresholding activation values

can mitigate pattern interference.

7.1.5 Hybrid Deep Learning + CMM System

A promising direction is to use deep models for feature extraction and CMM for lightweight storage and fast recall.

7.1.6 Robustness Tests

Evaluate performance under:

- Occlusion;
- Noise;
- Illumination changes; and
- Misalignment.

This would help understand operational boundaries.

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